

# VAIBHAV SINGH

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## Education

### Jaypee Institute of Information Technology

Bachelor of Technology, Computer Science and Engineering

Noida, India

Aug 2023 – May 2027

- CGPA: 7.11 / 10.0
- Coursework: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, Machine Learning

## Technical Skills

**Languages:** Python, C++, SQL

**Backend:** FastAPI, REST API design, JWT authentication, RBAC, SQLAlchemy ORM, PostgreSQL, MySQL

**Infrastructure and Tools:** Docker, Git, Render, Vercel, Supabase

**Machine Learning:** PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy

## Experience

### Drona Infotech

Software Engineer Intern (Backend)

Noida, India

Jun 2026 – Present

- Gave managers a running view of revenue owed versus collected, by building a billing module that stores each client's agreed fee and recomputes the outstanding balance whenever staff log a received payment.
- Blocked cross-role data access for 3 user types on a live HR platform, by implementing role-based access control (RBAC) in FastAPI that hashes passwords with bcrypt and binds each user's role to a signed JWT.
- Let the HR team run daily operations from one place, by building attendance and client-management modules on a normalized PostgreSQL schema modeled through SQLAlchemy ORM, that track check-ins and follow each client from onboarding to delivery.
- Made the full website run the same on any teammate's machine, by containerizing the service and its dependencies with Docker.

## Projects

### Academic Digital Twin — FastAPI, PostgreSQL, Docker, Scikit-Learn — Live [GitHub](#)

May 2026

- Supported every user flow across 18 REST endpoints covering auth, logging, analytics, simulation, and PDF export, by deploying a three-tier system: Vercel frontend, containerized FastAPI service on Render, and a Supabase PostgreSQL data layer.
- Cut feedback time from weeks to under a second, by building a what-if engine that rescores performance and stress on any change to study hours, sleep, or workload.
- Returned burnout, performance, and goal-alignment predictions in under 1 second from 5 daily inputs, by serving Random Forest and clustering models behind the API.

### Soul: Emotion and Mental Health Platform — PyTorch, Transformers, Docker

Feb 2025

- Reached 93% accuracy across 28 emotion classes and 0.928 F1 on crisis detection, by fine-tuning RoBERTa-base on 43K GoEmotions samples with early stopping over 3 epochs.
- Cut per-request latency for concurrent users, by loading the model and tokenizer once into memory at startup instead of on every request.
- Flagged every high-risk entry with no silent misses, by mapping each detected emotion to a clinical risk level and firing an alert whenever a score crossed the threshold.

### Plant Disease Classifier — TensorFlow, Keras, Docker — Live [GitHub](#)

Jun 2024

- Reached 93.9% test accuracy across 38 classes spanning 14 crops, by fine-tuning MobileNetV2 on the PlantVillage dataset with selective layer unfreezing.
- Kept the trained model at 13 MB so it runs on low-power devices, by using a MobileNetV2 backbone and training on 224x224 images with rotation, flip, and zoom augmentation.

## Research Publication

### Smart Retail: ML for Demand Prediction, Pricing and Inventory Management — IEEE CICN 2024

Dec 2024

- Beat classical forecasting baselines to 8.15% MSE on demand and 1.11% MSE on price across 10 products, by building a Random Forest pipeline on competitor pricing, customer preference, and seasonal features.
- Earned 4 citations and 532 full-text views, by presenting the paper at IEEE CICN 2024 (16th International Conference, Indore) after validating the model with stable 10-fold cross-validation.